01 / PROBLEM AND CONTRIBUTION

Folded-cascode amplifier

EEE 523 · Advanced Analog IC Design | Spring 2026 | Individual design

RESULT AT A GLANCE

Designed and simulated a folded-cascode/Class-AB amplifier with approximately 70 dB gain and 57.5° phase margin; a separate 1 pF simulation reached 169 MHz unity-gain frequency.

The engineering question

How can a folded-cascode gain stage drive a load through a Class-AB output buffer while preserving gain and stability?

Situation

A high-gain CMOS amplifier must also bias correctly and remain stable when its output stage drives a load. This project combines a folded-cascode gain stage with a Class-AB buffer.

Task

Size and bias the transistor-level design, then evaluate gain, stability, and load-dependent bandwidth.

My contribution

Transistor sizing, biasing, and simulation in the EEE 523 project report.

Tools and concepts

Analog IC design / CMOS / Cadence Virtuoso / Biasing / Stability analysis

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

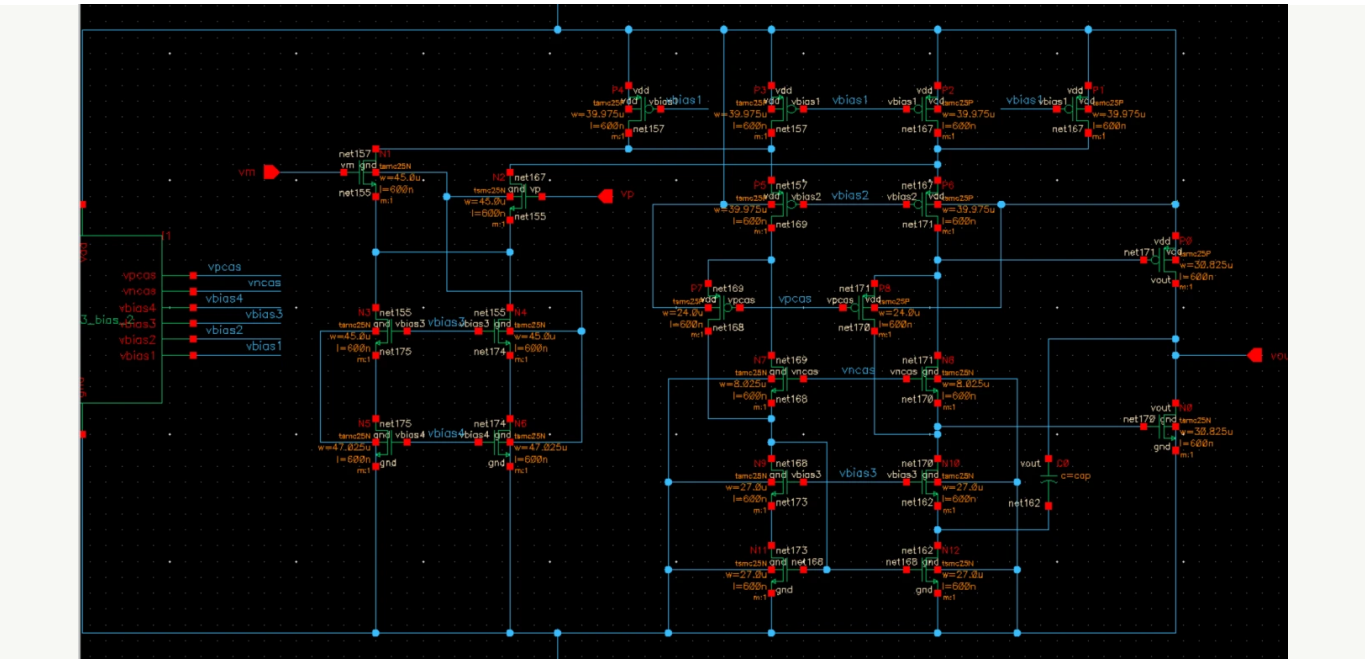
Combined a folded-cascode amplifier with a Class-AB output stage and sized the transistors around their bias conditions.

02 / Implementation step

Used operating-point and AC simulations to examine gain, phase margin, and unity-gain frequency. The study treats the load condition as part of each result rather than reporting a single unconditional bandwidth.

Bias and signal path

The bias network establishes device operating points before the folded-cascode stage is evaluated. The Class-AB buffer is a separate output-stage element, so the complete amplifier must be assessed with its load included.



Folded-cascode stage and Class-AB buffer

Transistor-level schematic from the amplifier report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

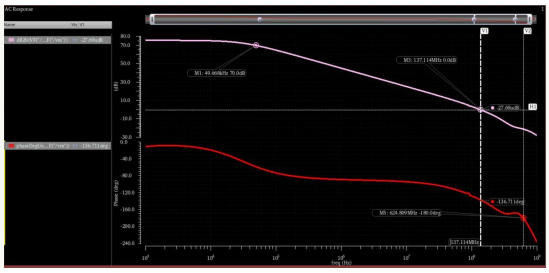
Open-loop gain	≈70 dB
Phase margin	57.5°
Unity-gain frequency	169 MHz

Reported result

The report and approved brief show approximately 70 dB open-loop gain and 57.5° phase margin. A separate 1 pF load simulation gives 169 MHz unity-gain frequency.

Reading the AC response

Open-loop gain, phase margin, and unity-gain frequency answer different questions. Keep the reported 57.5° stability reading separate from the 169 MHz result obtained with a 1 pF load; they are not a single universally valid operating specification.



Gain and phase response

Reported stability simulation; see the evaluation for load conditions and target comparison.

Scope and limitations

The 57.5° phase margin is below the 60° design target. It identifies stability as a remaining design consideration. The 169 MHz unity-gain result is from a separate simulation with a 1 pF load.

Possible next steps - proposed, not completed results

- Re-run gain, phase margin, and bandwidth under one explicitly documented load condition.
- Evaluate compensation and load dependence against the 60° phase-margin target.

Source material: EEE 523 Project 3 report, pp. 4, 9, 15, 21-22; approved amplifier brief.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.